

- It is great for simple, brute-force type of testing.

Official website: <https://www.joedog.org/>

Latest version: 4.0.4

Multi-Mechanize

Multi-Mechanize is an open source framework for performance and load testing. It runs concurrent Python scripts to generate load (synthetic transactions) against a remote site or service. Multi-Mechanize is most commonly used for Web performance and scalability testing, but can be used to generate workloads against any remote API accessible from Python.

Multi-Mechanize is pure Python and you have access to all of Python's standard libraries in your scripts. For example, you can use *httplib* to write a virtual user agent script and get detailed HTTP profiling times (TTFB, TTLB, etc).

Features:

- A powerful tool using Python.
- Enables running concurrent scripts to generate load.
- Generates detailed graphs for hits/throughput and concurrent requests.
- Generates detailed request/response debug files.
- Test output reports are saved as HTML or JMeter-compatible XML.

Official website: <https://multi-mechanize.readthedocs.io/en/latest/>

Latest version: 1.2.0

Apache Bench

Apache Bench (ab) is a load testing and benchmarking tool for a Hypertext Transfer Protocol (HTTP) server. It can be run from the command line and is very simple to use. The Apache Bench tool (AB) can load test servers by sending an arbitrary number of concurrent requests. Although AB was designed for testing Apache installations, it can be used to benchmark any HTTP server. It works by generating a flood of requests to a given URL, and returns some easily digestible performance related metrics to the screen. This simplicity makes it appealing for running quick and dirty load tests, and helps you discover the limitations of your Web stack or a service bottleneck that you did not anticipate.

Features:

- Multiple MPMs (multi-processing modules) can now be built as loadable modules at compile time.
- Less memory utilisation, asynchronous support and fully supported event MPMs.
- Smart filtering and SQL database support.

Official website: <https://httpd.apache.org/docs/2.4/programs/ab.html>

Latest version: 2.4

K6

K6 is an open source load and performance testing tool built using the Go and JavaScript languages. It provides a clean, approachable scripting API, local and cloud execution, flexible configuration, with command and control through CLI or a REST API. It can do tests ranging from simple unit load tests to complex load tests, and supports HTTP/1.1/2 and WebSockets.

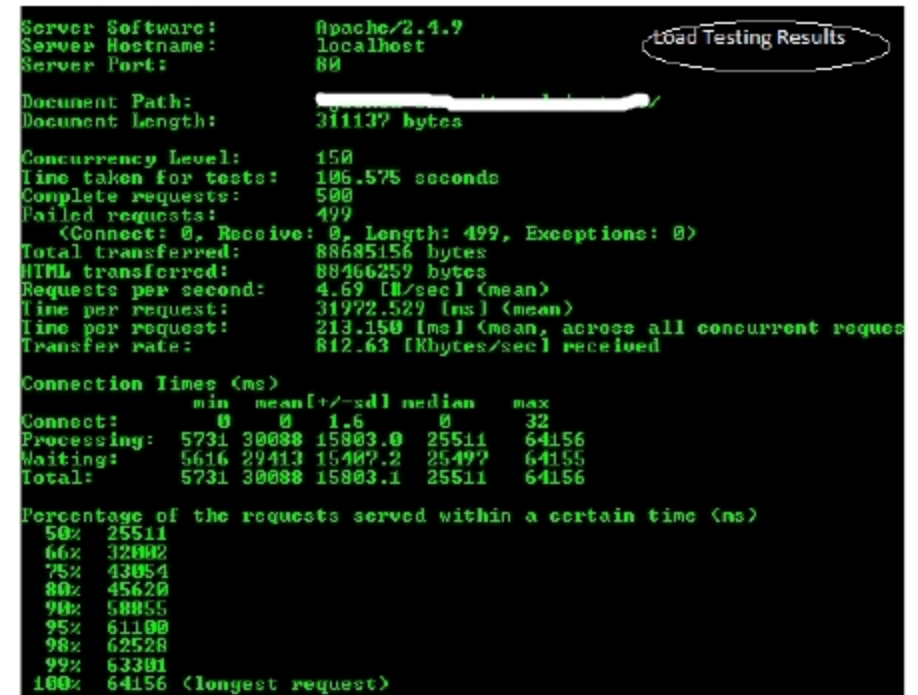


Figure 5: Apache Bench load testing results

Features:

- Provides easy and flexible configuration for testing.
- TLS features: Client certificates, configurable SSL/TLS versions and ciphers.
- Crypto library for implementing authentication/authorisation protocols.
- Built in the HAR converter and provides support for modules to aid code reusability.

Official website: <https://k6.io/>

Latest version: 0.23.1 

References

- [1] <https://locust.io/>
- [2] <https://github.com/httpperf/httpperf>
- [3] <https://jmeter.apache.org/>
- [4] <https://www.joedog.org/>
- [5] <https://multi-mechanize.readthedocs.io/en/latest/>
- [6] <https://httpd.apache.org/docs/2.4/programs/ab.html>
- [7] <https://k6.io/>

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